Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions.

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Dataset at a Glance

3,900

Purchases

18

Features

50

Locations

4

Categories

Demographics

Age, Gender, Location, Subscription

Purchase Details

Item, Category, Amount, Season, Size

Behavior

Discounts, Frequency, Ratings, Shipping

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Data Preparation in Python

01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Standardize

Imputed missing Review Ratings; renamed columns to snake_case

03

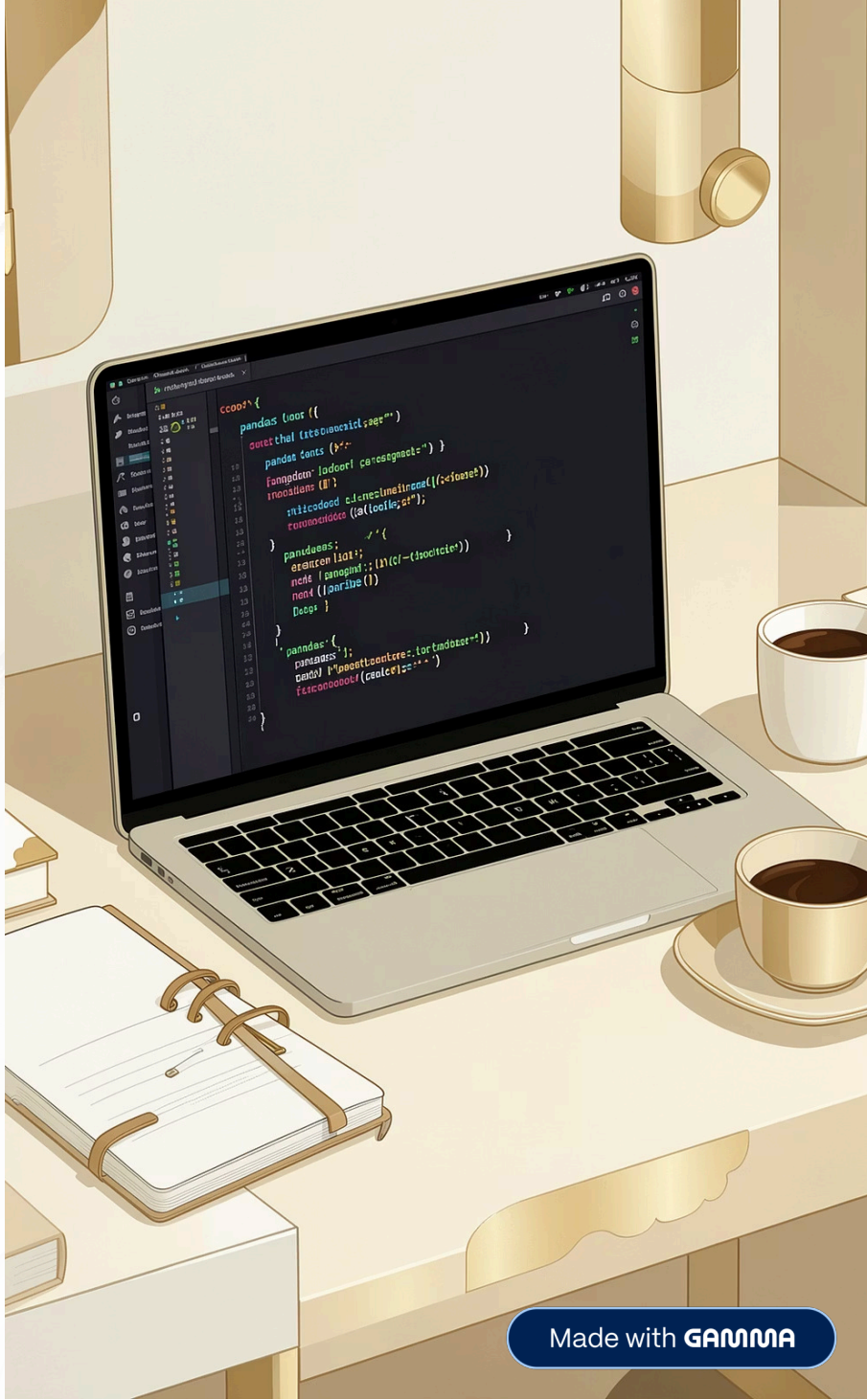
Feature Engineering

Created `age_group` and `purchase_frequency_days` columns

04

Database Integration

Loaded cleaned data into PostgreSQL for SQL analysis



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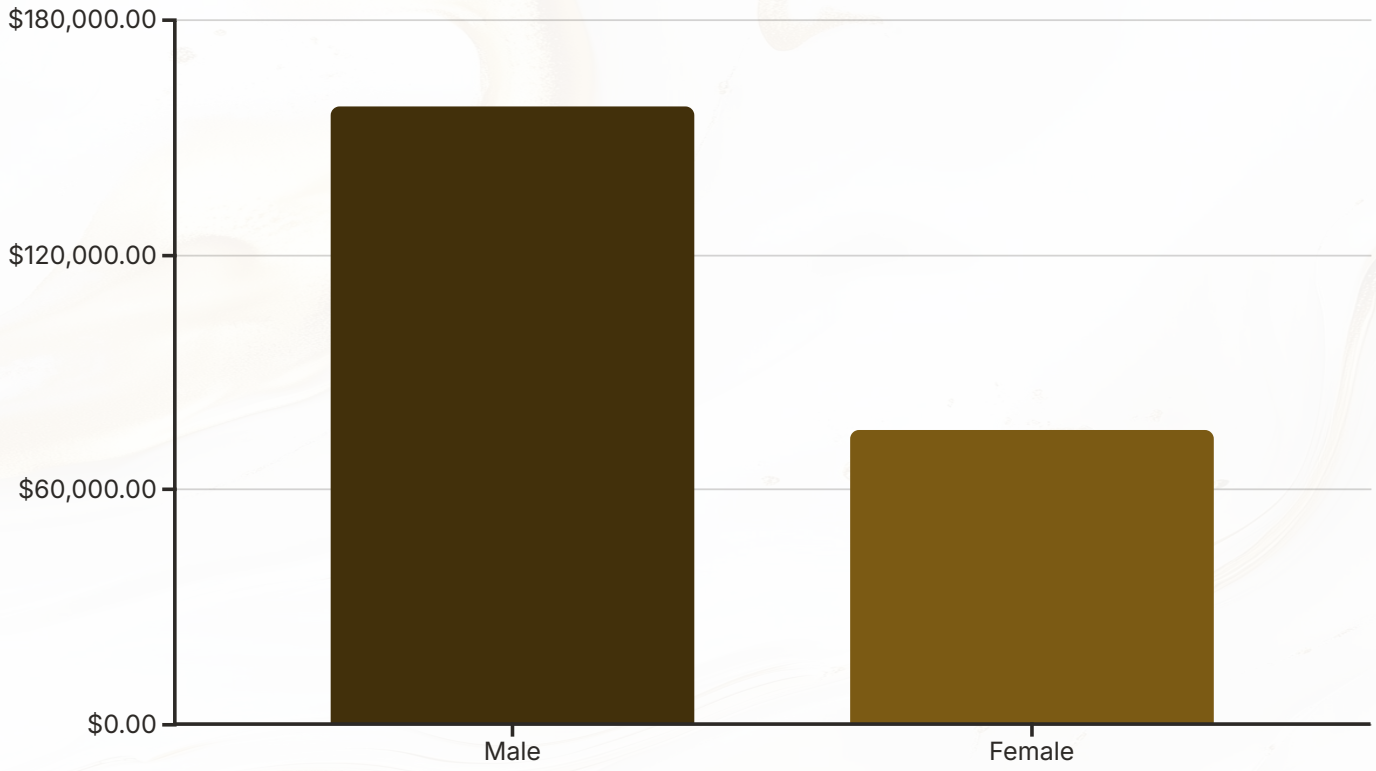
Revenue Insights



Revenue by Age Group

Young Adult	$62,143
Middle-aged	$59,197
Adult	$55,978
Senior	$55,763

Male customers generate **2× more revenue** than female. Young Adults lead by age group.

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Top-Rated Products & Shipping

Top 5 Products by Avg Rating

Result Grid | Filter Rows:

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Shipping Type vs. Avg Spend

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

shipping_type	ROUND(AVG(purchase_amount),2)
Express	60.48
Standard	58.46

Express shoppers spend **$60.48** vs. Standard at **$58.46**

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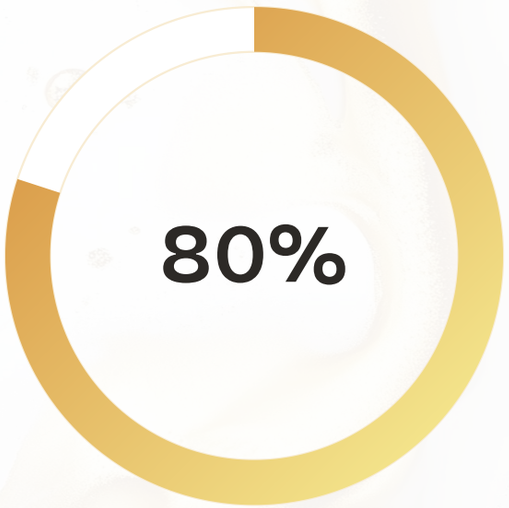
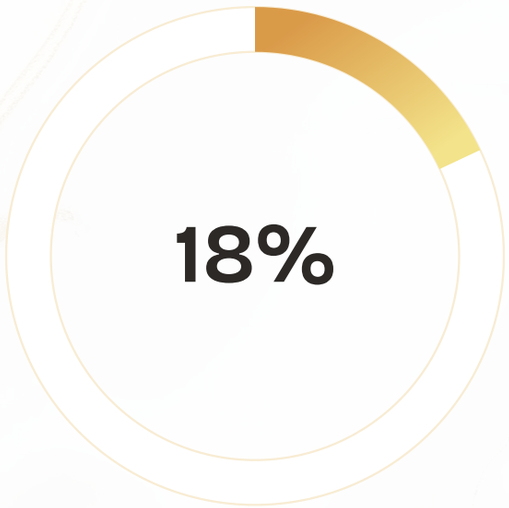
Customer Segmentation



Loyal Customers

3,116 customers



Returning

701 customers

New

83 customers

Repeat Buyers & Subscriptions

Customers with **>5 purchases**:

Subscribed	958
Not Subscribed	2,518

Only **27%** of customers subscribe — a major growth opportunity.

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Discount & Product Insights

Most Discount-Dependent Products

Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Top Products per Category

Accessories	Jewelry	171
Clothing	Blouse	171
Footwear	Sandals	160
Outerwear	Jacket	163

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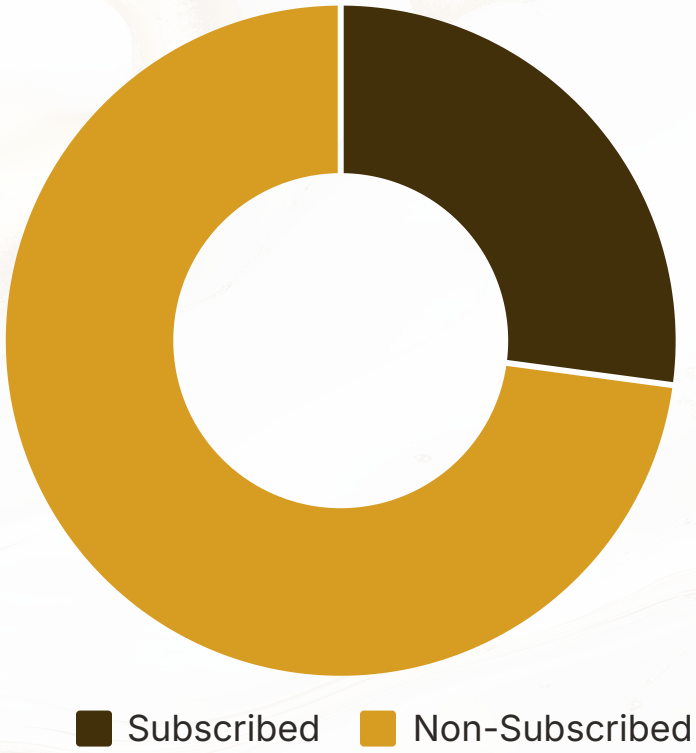
Subscribers vs. Non-Subscribers



Spend Comparison

Subscribed	Avg $59.49	Rev $62,645
Non-Subscribed	Avg $59.87	Rev $170,436

⚠ Non-subscribers drive **73%** of total revenue — converting them is critical.

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Power BI Dashboard

3.9K

Customers

$59.76

Avg Purchase

3.75

Avg Rating

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Business Recommendations

→ Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscribers

→ Loyalty Programs

Reward repeat buyers to grow the Loyal segment

→ Review Discount Policy

Balance sales boosts with margin control on high-discount items

→ Targeted Marketing

Focus on Young Adults and Express-shipping high-revenue users

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